

Risk-Aware Occupancy for Safety-Oriented End-to-End Autonomous Driving

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Abstract

Sparse representation formulates the environment perception for the end-to-end driving system as a set of discrete elements like objects and lane lines. This formulation meets safety risks in crowded, occluded scenes dealing with unstructured obstacles, uncertain regions, and intricate interactions. In this paper, we propose a dense representation, risk-aware occupancy, to characterize planning-relevant risks in an explicit and uniform manner. It jointly encodes global scene occupancy, map-derived traffic constraints, and future dynamic agent occupancy into a unified BEV map. The unified BEV map captures the risk evidence for trajectory planning in both spatial and temporal dimensions. We design an E2E network, ROIDrive, to realize risk-aware occupancy. It predicts risk-aware occupancy with an independent branch and injects it into planning queries for safety-oriented trajectory generation. In addition, to quantify the safety problem, we introduce RiskOcc4D-nuScenes built upon nusenes and occ3d-nusenes. Our risk-aware occupancy yields relative open-loop collision reductions of **52.9%** under the UniAD metric and **35.0%** under the ST-P3 metric on nuScenes.

Introduction

Safety is one of the core problems blocking the wide application of both L2 and L4 autonomous driving. Current end-to-end methods have made great progress on the safety problem, but the sparse representation makes it hard to completely and explicitly tackle the planning-relevant risks. When the environment is encoded into discrete agents and maps elements, a natural safety risk arises in dealing with unstructured elements and occluded regions. In contrast, a dense representation could describe the scene as a continuous spatial field. Then the planner can explicitly check whether candidate trajectories are feasible, and measure the conflict risk around them. In this paper, we propose risk-aware occupancy as the dense representation and make it flexible to help existing e2e frameworks become safety-oriented.

By diving into the detailed scenes of safe driving, we summarize two key attributes for the dense representation. As illustrated in Figure 1, first, it should continuously distinguish drivable from non-drivable space, which has an arbitrary shape, in the time dimension. Then, it should evaluate spatial conflicts by predicting how these conflicts evolve over time. These insights follow the intuitive human driving principles, where a driver will determine whether the surround

environment could let the vehicle safely pass before considering the fine-grained categories. (Mecheri, Mars, and Lobjois 2023; Papakostopoulos, Marmaras, and Nathanael 2017). For example, a relatively occupied height indicates whether a region is physically drivable for safety. A map structure could define the admissible motion space in law and avoid dangerous driving. Furthermore, the future motion could reveal emerging conflicts to prevent possible collisions. After that, fine-grained semantic information could help deal with traffic rules and hazard severity (Tavakoli et al. 2023; Tavakoli and Heydarian 2022). Therefore, we propose the risk-aware occupancy map as the dense perception representation, which could explicitly model 3D geometry and spatial interaction and be interpretable for safety inspection. It encodes the drivable area, spatial conflicts, and future motion into one unified 3d space. For safety-oriented driving, it also directly computes the planning-relevant risk beyond the geometric and semantic information.

Inspired by (Chen et al. 2024). The risk-aware occupancy is built on a unified bev risk field, where each cell could model a comparable risk score. The risk score is computed from object geometry, safety-relevant semantic priors, map constraints, and future motion cues. With the unified field, the ego vehicle can encounter risk directly queryable by the planner. Furthermore, projecting planning-relevant 3D structure into BEV reduces redundant vertical representation and matches the spatial feature layout used by modern E2E networks. Therefore, the risk-aware occupancy can serve both as a supervisory target for perception and as an intermediate feature for planning. It aligns the perception objective with downstream decisions and provides a direct interface for safety inspection, failure localization, and visual debugging.

We design a risk-aware occupancy enhanced instance-centric E2E (ROIDrive) to prove the priority of risk-aware occupancy for an end-to-end framework. The risk-aware occupancy is predicted by an independent branch and fused into planning. The integration is seamless for any kind of end-to-end network design by replacing the sparse representation with our risk-aware occupancy map. Besides, to ensure efficient performance, we also tokenize the dense risk field by sliding-window sampling and inject it into the agent queries via cross-attention, staying orthogonal to the sparse tokens.

To fully study the safety problem, we also present a new dataset *RiskOcc4D-nuScenes* from Occ3D-nuScenes. The

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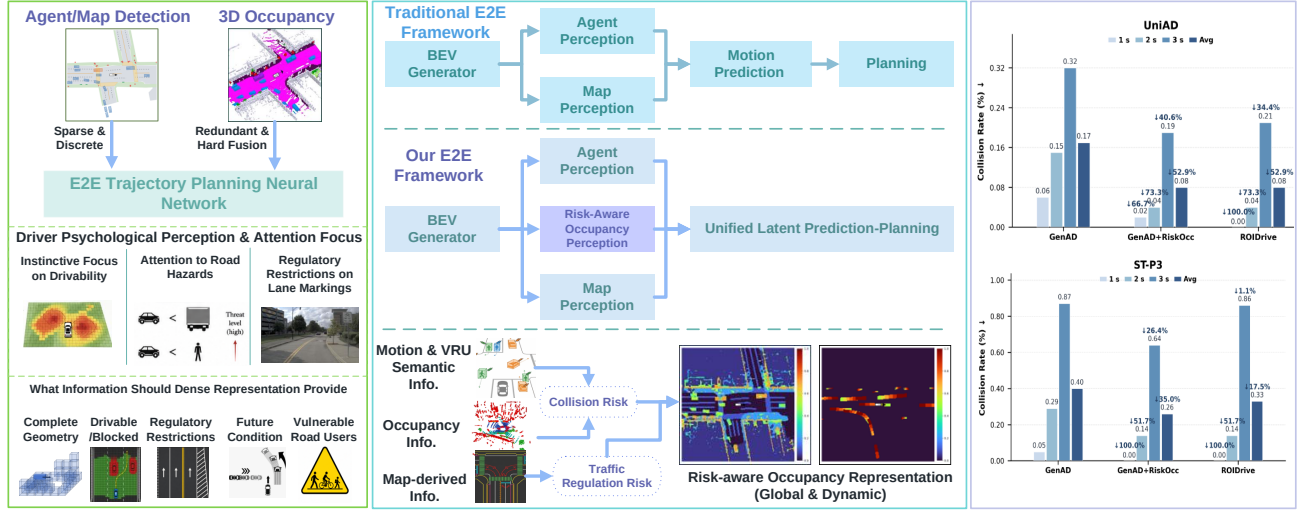


Figure 1: Motivation and design of risk-aware occupancy. Left: Limitations of sparse agent/map detection and naive 3D occupancy, and human-inspired requirements for dense representations covering geometry, drivability, traffic rules and vulnerable road users. Middle: Comparison between conventional separated E2E pipelines and our unified prediction-planning framework, which constructs global dynamic risk-aware occupancy by encoding collision and traffic regulation risks. Right: Collision-rate benchmarks on UniAD and ST-P3 metrics, verifying the safety improvement of the proposed risk-aware occupancy representation.

dataset is built through four automated annotation pipelines providing a reliable risk coverage under occlusion and distant point-cloud sparsity, which incorporates road-related and temporal cues. We conduct ablation on multiple 3D occupancy architectures to predict risk-aware occupancy, establishing a benchmark for representation learnability and model compatibility. By only replacing the oracle risk maps with our predicted risk-aware occupancy, we can separate the intrinsic value of the representation from prediction error and validate its practical utility for downstream planning.

The main contributions of this work are summarized as follows:

1. **Risk-aware occupancy representation.** We introduce a safety-oriented dense BEV representation that unifies geometry, safety-relevant semantics, map constraints, and future motion into an explicit planning-relevant risk field.
2. **ROIDrive E2E framework.** We design ROIDrive, an E2E driving framework with an independent risk-aware occupancy branch seamlessly integrated at the planning stage, and validate it throughout the complete perception-to-planning pipeline.
3. **RiskOcc4D-nuScenes dataset.** We construct the RiskOcc4D-nuScenes dataset using four automated annotation pipelines and instantiate risk-aware occupancy in the end-to-end framework to verify its effectiveness and quantify its performance improvements.

Related Work

Occupancy Prediction Methods

Recently, occupancy prediction has advanced rapidly along three axes. **Lightweight architectures:** FlashOcc (Yu et al.

2023), TPVFormer (Huang et al. 2023), SparseOcc (Tang et al. 2024), FastOcc (Hou et al. 2024), and Octree-Occ (Lu et al. 2023) reduce cost via BEV extraction, tri-plane decomposition, sparse encoding, and octree reconstruction, respectively. **Performance optimization:** FB-Occ (Li et al. 2023b) fuses forward-backward projection (Huang et al. 2021; Li et al. 2022) with multi-task learning; F-OCC (Zhao, Zheng, and Gong 2024) extends DCNv4 (Xiong et al. 2024) to 3D; OccTransformer (Liu et al. 2024), OccFormer (Zhang, Zhu, and Du 2023), VoxelFormer (Li et al. 2023a), and MambaOcc (Tian et al. 2024) leverage advanced attention or state-space architectures. **Cross-domain integration:** NeRF-based methods (Pan et al. 2023; Zhang et al. 2023; Huang et al. 2024b; Mildenhall et al. 2021), Gaussian splatting variants (Huang et al. 2024c,a; Gan et al. 2024; Kerbl et al. 2023), world models (Zheng et al. 2025), and LLM-driven approaches (Xu et al. 2025; Wei et al. 2024) further enrich occupancy representations from complementary perspectives.

Auxiliary Perception Supervision in End-to-End Driving Models

Recent end-to-end driving models widely adopt multi-task perception auxiliary losses to regularize shared BEV features. Typical branches include BEV 2D/3D detection, semantic segmentation, vectorized lane prediction, and dense 3D/4D occupancy forecasting. For instance, VADv2 (Jiang et al. 2026) and DiffusionDriveV2 (Zou et al. 2025) use BEV detection to constrain agent localization; HENet++ (Xia et al. 2025) adds map segmentation for lane-compliant trajectories; and OccVLA (Liu et al. 2025) and IR-WM (Mei et al. 2025) exploit volumetric occupancy for vertical obstacle ge-

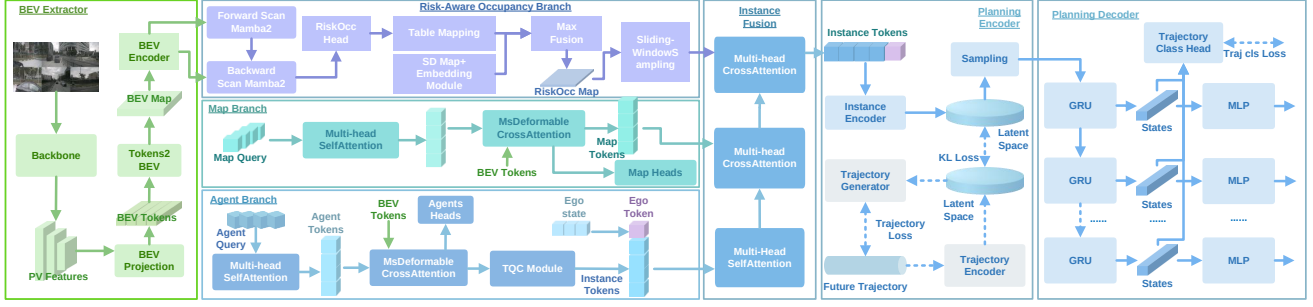


Figure 2: ROI Drive architecture. A BEVFormer backbone yields the BEV feature B ; the risk-aware occupancy (RiskOcc) branch regresses a normalized risk field via a Bi-Mamba2 block and a class-to-risk lookup table (optionally fused with an SDMap+ prior), tokenizes it by sliding-window sampling, and injects it into the queries via cross-attention. Temporal Query Consistency (TQC) then gates the queries by cross-frame reliability before a VAE generative planner decodes multimodal trajectories.

ometry, whereas the lightweight DrivR (Kirby et al. 2026) drops dense occupancy to cut cost at the expense of spatial risk cues. Yet all such supervision optimizes geometric or semantic objectives decoupled from trajectory ranking, and full 4D occupancy remains costly. We instead propose risk-aware occupancy, a compressed BEV risk signal that embeds trajectory-consistent spatiotemporal risk at low computational cost.

Risk-Aware Occupancy Perception

Conventional 3D semantic occupancy builds high-entropy voxel representations $H(\mathcal{V})$ with fine-grained categories; though complete, they carry substantial task-irrelevant redundancy $H_{\text{red}}(\mathcal{V})$ that degrades risk-information transfer for E2E driving.

Risk-aware occupancy prediction is formulated as planning-oriented minimal sufficient statistic extraction:

$$R^* = \arg \max_R I(R; T_{\text{plan}}), \quad \text{s.t.} \quad \dim(R) \ll \dim(\mathcal{V}), \quad (1)$$

where $I(R; T_{\text{plan}})$ is the mutual information between risk representation R and the optimal trajectory T_{plan} .

Moreover, conventional 3D occupancy is over-complete: its semantic paradigm cannot uniformly quantify collision threats and regulatory constraints (e.g., lane markings, stop lines). Risk-aware occupancy encodes both risk types and resolves conflicts via maximum aggregation over collision and regulatory terms. To realize this formulation, dual-axis reduction is performed:

- **Spatial:** Only the critical height interval $H_{\text{crit}} \in [0 \text{ m}, 2.2 \text{ m}]$ (the vehicle-relevant collision zone) is retained, discarding non-threatening airspace.
- **Semantic:** Fine-grained labels are replaced by object height as the primary risk quantifier; vulnerable road users (VRUs) retain semantic priors with elevated risk weights.

A hierarchical risk quantification framework is constructed over the Occ3D-nuScenes perception domain $\Omega = \{(x, y) \mid x, y \in [-40 \text{ m}, 40 \text{ m}]\}$. Critical heights are nonlinearly mapped to a normalized risk score $Z_{\text{norm}} \in [0, 1]$. To separate risk modes, static global obstacle risk is bounded

to $V_{\text{risk}}^{\text{global}} \in [0.2, 0.7]$ and moving objects to a dedicated branch with $V_{\text{risk}}^{\text{dyn}} \in [0.5, 1.0]$. Their deliberate overlap enforces a strict priority hierarchy—dynamic collision > static obstacle > regulatory risk—so dynamic threats receive planning priority while gradient conflict in multi-task optimization is mitigated.

ROI Drive Framework

Design rationale. As illustrated in Figure 2, ROI Drive builds on the query-based generative E2E paradigm (Zheng et al. 2024; Jiang et al. 2023), where a BEVFormer backbone yields a BEV feature $B \in \mathbb{R}^{C \times H \times W}$, a small fixed set of agent and map queries abstracts the scene, and a conditional VAE planner decodes multimodal trajectories. This compression causes two structural weaknesses: *representation sparsity*, as the dense, temporally evolving risk structure of the BEV plane is left implicit, and *token sparsity and unreliability*, as missed detections erase risk while false-positive and flickering queries inject spurious motion intent. ROI Drive thus pursues two goals: (1) to introduce a perception modality that is independently predicted and then fused into the E2E pipeline, and (2) to densify the representation and calibrate token reliability. Both are met by two modules at the motion-prediction stage: a risk-aware occupancy (RiskOcc) branch and a Temporal Query Consistency (TQC) module.

Risk-aware occupancy branch. The BEV feature is channel-reduced, modeled by a Bi-Mamba2 state-space block that captures long-range dependencies at $\mathcal{O}(N)$ cost, and upsampled. A lookup table converts the stable class predictions to a normalized risk field $R \in [0, 0.7]$; after optional fusion with an SDMap+ prior, R is tokenized by sliding-window sampling and injected into the queries via cross-attention. The field is dense and orthogonal to the queries: the queries encode object identity and location, while the field encodes the current and near-future risk at each location. It also provides a detection-free fallback for missed hazards, since R is regressed directly from B , and anticipates risk through long-range scanning and future supervision.

Temporal Query Consistency. TQC calibrates query reliability before motion self-attention without learnable parameters. It caches previous-frame agent states, aligns them

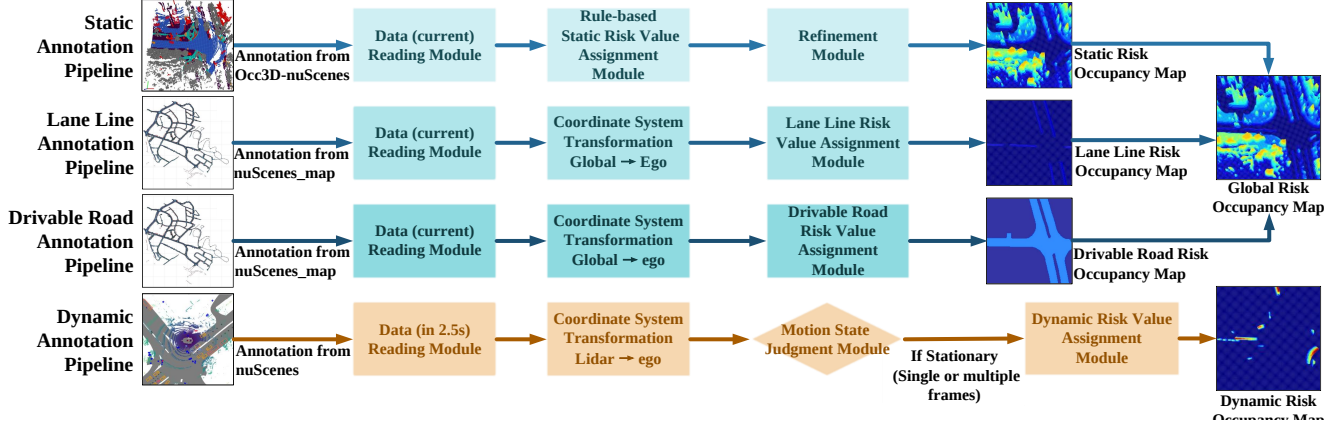


Figure 3: The four automated annotation pipelines building RiskOcc4D-nuScenes, fusing static, regulatory, drivable-area, and dynamic-future cues into the global and dynamic risk-aware occupancy maps.

to the current ego frame via CAN-bus ego-motion, and matches each query to its nearest predecessor. A consistency weight fuses feature cosine similarity with a distance-decayed spatial score, with a fallback for newly appearing objects and a lower bound against hard deletion. The weights form a pairwise geometric-mean gate mapped to a log-odds bias β that modulates attention through a stop-gradient $\text{sg}(\cdot)$: $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d} + \text{sg}(\beta))V$. Consistent tokens are amplified and spurious or flickering ones driven toward zero attention; being inference-only and gradient-free, TQC is lightweight and falls back to the base model on the first frame.

Generative planning. The instance tokens, densified by RiskOcc, reweighted by TQC, and initialized at the ego slot from historical ego states (Li et al. 2023b), are encoded by a VAE into a latent Gaussian $p(z | T) \sim \mathcal{N}(\mu_i, \sigma_i)$ (Zheng et al. 2024); a sampled z is decoded by a GRU and an MLP into multimodal ego and agent trajectories.

RiskOcc4D-nuScenes Dataset

Conventional 3D occupancy datasets suffer from spatial redundancy, semantics misaligned with E2E planning, the absence of a unified collision-regulatory risk encoding, and missing temporal risk annotations. We address these limitations with *RiskOcc4D-nuScenes*, built upon nuScenes and Occ3D-nuScenes—the first 4D risk-aware occupancy benchmark for E2E autonomous driving. Four automated pipelines unify static global risk, traffic regulatory risk, and dynamic future risk into a single framework while remaining fully compatible with existing 3D occupancy algorithms.

Task Definition

Combining the avoidance-priority principle with Bayesian risk decision theory, we instantiate the abstract optimal representation R^* in Eq. (1) by minimizing the bottleneck risk that aligns perception with planning:

$$\mathcal{L}(T_{\text{plan}} | Z) = \mathbb{E}[\max(\ell_{\text{col}}(T_{\text{plan}}, O), \ell_{\text{reg}}(T_{\text{plan}}, M)) | Z], \quad (2)$$

where Z is the multi-view input, (ℓ_{col}, O) the collision risk and obstacle set, and (ℓ_{reg}, M) the regulatory risk and traffic-rule constraints. The max operator enforces hard-constraint alignment, avoiding accumulation overflow and inter-term gradient conflict, thereby providing the supervisory signal for the risk-aware occupancy maps in Section .

Input. Six surround-view camera sequences under nuScenes calibration: intrinsics $\{K_i\}$, extrinsics $\{[R_i | t_i]\}$, and N consecutive frames $\{I_{i,t} \in \mathbb{R}^{H_i \times W_i \times 3}\} (i=1, \dots, 6; t=1, \dots, N)$.

Output. Ego-centric BEV risk-aware occupancy maps: a global map $RM_t \in \mathbb{R}^{H \times W}$, where occupancy encodes spatial location and risk values quantify hazard attention carrying collision and regulatory information, and a dynamic map RM_d that fuses future steps Δt to encode motion-prediction uncertainty. They provide three targets: (1) discrete global risk classification, (2) continuous global risk regression, and (3) dynamic temporal risk prediction.

Dataset Construction Pipelines

Our four-stage automated framework generates the global and dynamic risk-aware occupancy maps (Figure 3), compressing redundant 3D representations while enriching label completeness.

Static Annotation Pipeline (Pipeline-1) This pipeline extracts static risk priors from Occ3D-nuScenes via 3D-to-BEV reduction. For each x - y cell, vertical marginalization within the critical height range $H_{\text{crit}} \in [0, 2.2]$ m projects semantic and geometric cues onto the BEV plane, where the driving risk attention V_{risk} is a joint function of local height distribution and semantic category priors. To suppress quantization artifacts and smooth jagged boundaries, we refine the initial risk field $\mathbf{R}_{\text{init}}$ via morphological dilation-erosion and a custom convolution-kernel infilling: $\hat{\mathbf{R}}_0 = \mathcal{F}((\mathbf{R}_{\text{init}} \oplus \mathbf{K}_d) \ominus \mathbf{K}_e)$, where $\oplus$ and $\ominus$ denote dilation and erosion with structuring elements $\mathbf{K}_d$ and $\mathbf{K}_e$, and $\mathcal{F}(\cdot)$ propagates neighboring context to repair sparse regions.

Table 1: Feasibility benchmark for predicting risk-aware occupancy with representative 3D occupancy backbones.

Baselines	Release	Metrics						
		Global					Dynamic	Infer. Speed
		mIoU $\uparrow$	MAE $\downarrow$	MAE@15 $\downarrow$	MAE@30 $\downarrow$	SSIM $\uparrow$	Mask-MAE $\downarrow$	FPS $\uparrow$
PanoOcc-S (Wang et al. 2024)	CVPR 24	14.54	0.121	0.022	0.079	0.131	0.732	6.0
PanoOcc (Wang et al. 2024)	CVPR 24	14.71	0.121	0.022	0.079	0.131	0.731	1.9
OccProphet (Chen et al. 2025)	ICLR 25	17.39	0.099	0.017	0.067	0.355	0.524	0.7
BEVDet4DOcc (Huang and Huang 2022)	-	19.71	0.093	0.015	0.062	0.373	1.856	9.5
ViewFormerOcc (Li et al. 2024)	ECCV 24	21.38	0.093	0.015	0.060	0.379	0.734	19.0
FO(BEVDet) (Yu et al. 2023)	ICCV 23	22.03	0.088	0.014	0.058	0.424	0.682	66.2
BEVFormer-Occ (Li et al. 2022)	ECCV 22	23.76	0.086	0.014	0.057	0.437	0.565	11.1
CVTOcc (Ye et al. 2024)	ECCV 24	24.43	0.094	0.016	0.063	0.385	0.623	2.1
FO(Stereo4D) (Yu et al. 2023)	ICCV 23	25.14	0.082	0.011	0.052	0.443	0.664	10.4
DHDOcc-S (Wu et al. 2025)	ICRA 25	25.45	0.092	0.015	0.061	0.399	0.610	18.7
ProtoOcc (Kim et al. 2025)	AAAI 25	28.58	0.082	0.012	0.053	0.422	0.589	7.1
FB-Occ (Li et al. 2023b)	ICCV 23	32.96	0.075	0.010	0.048	0.480	0.675	34.1
DHDOcc-M (Wu et al. 2025)	ICRA 25	34.15	0.079	0.011	0.050	0.455	0.559	3.5

Lane Line Annotation Pipeline (Pipeline-2) This pipeline extracts an ego-centric $80\text{m} \times 80\text{m}$ semantic map from nuScenes and injects regulatory priors by assigning $V_{\text{risk}} = 0.1$ to road-marking cells (crosswalks, stop lines, lane dividers). This explicit rule encoding prunes the trajectory hypothesis space, reduces decision entropy, and sets a low-risk regulatory baseline in the hierarchical risk framework.

Drivable Road Annotation Pipeline (Pipeline-3) To counter missing drivable-area labels and LiDAR-sparsity noise in Occ3D-nuScenes, high-fidelity drivable masks from nuScenes HD maps are fused with the Pipeline-1/2 outputs through deterministic spatial fusion, completing undefined areas and removing outliers (Figure 3). For occluded blind-spot cells with unobservable occupancy, a non-informative prior of 0.05—below regulatory and collision risk yet strictly positive—encodes epistemic uncertainty, guiding cautious attention toward unobserved regions until later viewpoints resolve a definite ground truth. Fusing Pipelines 1–3 yields the global risk-aware occupancy map, stored as both classification and regression labels.

Dynamic Annotation Pipeline (Pipeline-4) This pipeline builds the dynamic risk potential field from observable temporal occupancy rather than unreliable kinematic fitting. Moving objects ($v > 0.5\text{m/s}$) are tracked across frames to record actual occupied positions over the next 2.5 s at 0.5 s intervals. Risk attention is inversely proportional to the time step—near-future moments carry higher attention, following the “urgent-near, relaxed-far” defensive-driving logic—yielding an initial field $\hat{\mathbf{R}}_0^{\text{dyn}}$. A spatial Gaussian diffusion $\mathbf{R}^{\text{dyn}} = \mathcal{G}_\sigma * \hat{\mathbf{R}}_0^{\text{dyn}}$ (kernel $\mathcal{G}_\sigma$, * spatial convolution) produces a continuous field with Lipschitz-continuous variation, modeling risk attenuation toward free space. Per Section ,

dynamic objects take $V_{\text{risk}}^{\text{dyn}} \in [0.5, 1.0]$ and are stored separately from static labels to avoid multi-task interference.

Experiments

We evaluate multiple 3D occupancy backbones on RiskOcc4D-nuScenes and its downstream utility for E2E driving.

Experimental Setup

Implementation details All occupancy baselines use their default configurations with only the output heads replaced: the classification head retains each method’s original supervision, while both regression heads use the Huber loss (Huber 1964). Training runs on multiple NVIDIA A100 GPUs; inference and benchmarking use an Intel i9-13900K CPU with an NVIDIA RTX 4090 GPU.

Evaluation Metrics We adopt a four-fold protocol. (1) *Classification* uses mIoU (mean Jaccard index) (Everingham et al. 2010) for risk-category alignment; (2) *Regression* uses MAE (L_1), with MAE@15/MAE@30 restricted to ego-centric 15×15 and $30 \times 30\text{m}^2$ regions to emphasize the safety-critical near field; (3) *Structural fidelity* uses SSIM (Wang et al. 2004) for luminance, contrast, and structural consistency beyond pointwise accuracy. (4) *Dynamic prediction* uses Mask-MAE, which counters the extreme imbalance from dominant background zeros via a binary mask $M_i = \mathbf{1}[P_i > \tau \text{ or } G_i > \tau]$ ($\tau = 0.1$):

$$\text{Mask-MAE} = \frac{1}{\sum_i M_i} \sum_{i=1}^N M_i |P_i - G_i|, \quad (3)$$

restricting evaluation to the support $\max(P_i, G_i) > \tau$ and improving discriminability.

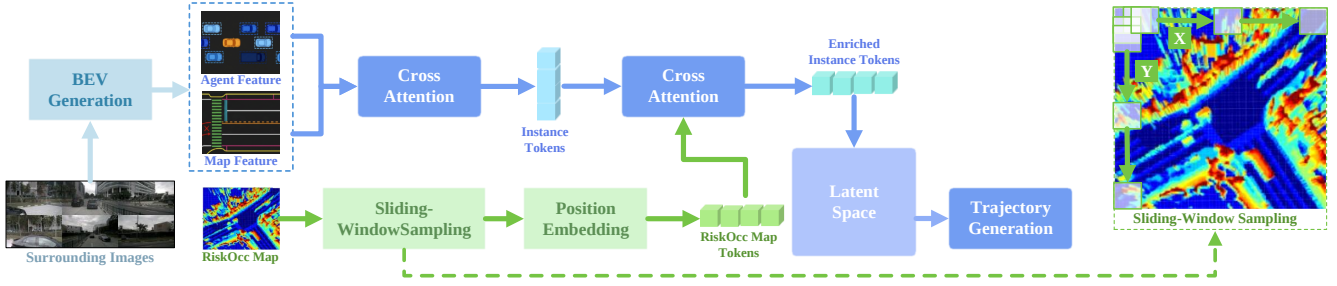


Figure 4: Architecture of the RiskOcc-GenAD fusion experiment: the predicted risk-aware occupancy map is sampled by a sliding window and fused with GenAD’s instance tokens via cross-attention, entering as an additional input while GenAD’s protocols stay unchanged.

Benchmarking on RiskOcc4D-nuScenes

We migrate a broad set of 3D occupancy methods to RiskOcc4D-nuScenes—PanoOcc (Wang et al. 2024), OccProphet (Chen et al. 2025), BEVDet4DOcc (Huang and Huang 2022), ViewFormerOcc (Li et al. 2024), FlashOcc(FO) (Yu et al. 2023), BEVFormerOcc (Li et al. 2022), CVTOcc* (Ye et al. 2024), ProtoOcc (Kim et al. 2025), FB-Occ (Li et al. 2023b), and DHDOcc (Wu et al. 2025)—with results in Table 1.

Table 2: Collision rate (%) comparison among E2E methods. Ours = GenAD (Zheng et al. 2024) + RiskOcc Map. P. = progressive training.

Methods	ST-P3 Metrics ↓				UniAD Metrics ↓			
	1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.
ST-P3	0.23	0.62	1.27	0.71	-	-	-	-
OccNet	0.21	0.59	1.37	0.72	-	-	-	-
VAD-Tiny	0.21	0.35	0.58	0.38	-	-	-	-
VAD-Base	0.07	0.17	0.41	0.22	0.01	0.30	0.95	0.42
FusionAD	0.25	0.13	0.25	0.21	-	-	-	-
UniAD	-	-	-	-	0.05	0.17	0.71	0.31
PPAD (P.)	0.08	0.12	0.38	0.19	0.02	0.20	0.93	0.38
PPAD	0.07	0.15	0.36	0.19	0.03	0.22	0.73	0.33
PARADrive	-	-	-	-	0.07	0.25	0.60	0.30
AD-MLP	0.17	0.18	0.24	0.20	-	-	-	-
GenAD	0.05	0.29	0.87	0.40	0.06	0.15	0.32	0.17
Ours	0.00	0.14	0.64	0.26	0.02	0.04	0.19	0.08

Global risk-aware occupancy. DHDOcc-M (Wu et al. 2025) leads in mIoU (34.15), indicating strong risk-category discrimination, while FB-Occ (Li et al. 2023b) attains the best regression at all scales (MAE 0.075, MAE@15 0.010, MAE@30 0.048) and the highest SSIM (0.480), best preserving the ground-truth spatial gradients.

Dynamic risk-aware occupancy. DHDOcc-M achieves the best Mask-MAE (0.559) via its decoupled temporal architecture, with BEVFormerOcc close behind (0.565), underscoring the value of temporal attention. BEVDet4DOcc is an outlier (1.856), likely from distribution shift in its dynamic head; all others fall within 0.559–0.734. The generally limited performance highlights the challenge of multi-step risk

propagation.

Inference efficiency. FO(BEVDet) (Yu et al. 2023) reaches 66.2 FPS without temporal history but with limited accuracy, whereas FB-Occ (Li et al. 2023b) offers the best speed-accuracy trade-off (34.1 FPS) alongside the strongest global metrics.

Table 3: Ablation of intermediate representations on collision rate (%). Best in **bold**.

Inputs	ST-P3 Metrics ↓				UniAD Metrics ↓			
	1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.
RiskOcc Map	0.00	0.14	0.64	0.26	0.02	0.04	0.19	0.08
3D-Occ	0.08	0.29	0.74	0.37	0.09	0.18	0.33	0.20
Binary Map	0.14	0.39	0.90	0.48	0.11	0.21	0.39	0.24
Gaussian Map	0.27	0.51	0.86	0.55	0.21	0.32	0.37	0.33

End-to-End Autonomous Driving Integration

Experimental Design We integrate the risk-aware occupancy map into GenAD (Zheng et al. 2024), an E2E planner encoding dynamic agents and road topology as instance and map tokens. Such discrete object-level representations are spatially incomplete, capturing only detected targets while lacking explicit encoding of dense spatial risk, long-tail obstacles, and ambiguous occupied regions—gaps that our dense, continuous risk distribution complements. Let $\mathcal{Z}_{\text{inst+map}}$ and $\mathcal{Z}_{\text{risk}}$ denote the joint instance-and-map and the risk representations. Our central hypothesis is a positive conditional mutual information gain:

$$\Delta I = I(\mathcal{A}_{\text{safe}}; \mathcal{Z}_{\text{risk}} | \mathcal{Z}_{\text{inst+map}}) > 0, \quad (4)$$

i.e., the risk representation carries safety-relevant information not captured by instance and map tokens. The risk map is sampled via a sliding window and fused with GenAD’s instance tokens through cross-attention (Figure 4), allowing each instance token to attend to surrounding grid-level risk. GenAD’s original training and evaluation protocols remain unchanged and the risk map enters solely as an additional input, ensuring all gains are attributable to the risk representation.

Table 4:ROIDrive ablation 1. M1: Risk-Aware Occupancy Perception; M2: ego-token prior initialization; M3: TQC (Temporal Query Consistency).

Methods			UniAD Metrics								ST-P3 Metrics							
M1	M2	M3	L2 (m) ↓				Collision Rate (%) ↓				L2 (m) ↓				Collision Rate (%) ↓			
			1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.
✓			0.29	0.51	0.78	0.52	0.08	0.07	0.21	0.12	0.38	0.85	1.52	0.92	0.07	0.07	0.64	0.26
✓	✓		0.16	0.31	0.56	0.34	0.08	0.10	0.27	0.15	0.21	0.58	1.22	0.67	0.08	0.16	0.84	0.36
✓	✓	✓	0.16	0.32	0.57	0.35	0.00	0.04	0.21	0.08	0.21	0.59	1.26	0.69	0.00	0.14	0.86	0.33

Comparative Results Table 2 reports collision rates on the nuScenes validation set. Ours substantially reduces collisions over GenAD under both metrics: ↓52.9% on UniAD (Hu et al. 2023) (0.17% → 0.08% Avg.) and ↓35.0% on ST-P3 (Hu et al. 2022) (0.40% → 0.26% Avg.), consistently across all horizons and largest at the 2 s horizon (UniAD ↓73.3%; ST-P3 ↓51.7%), where accumulated trajectory uncertainty is high and dense spatial risk priors are most valuable. Ours also attains zero collisions at 1 s under ST-P3 and the second-lowest under UniAD (0.02%), achieving the lowest average collision rate under the UniAD metric among all compared methods.

Ablation Study Each ablation replaces the RiskOcc Map with an alternative spatial representation while keeping the fusion module and all other settings fixed (Table 3).

3D-Occ. Conventional 3D semantic voxels are adaptively pooled to the RiskOcc Map size and fused identically. Gains are mixed—slightly better than baseline under ST-P3 (0.37 vs. 0.40) but worse under UniAD (0.20 vs. 0.17): although raw voxels carry high mutual information, adaptive pooling lowers the effective information rate and planning-irrelevant semantic redundancy adds noise. By distilling safety-critical information at construction time, RiskOcc attains higher task-relevant density in a lower-dimensional BEV, with consistent gains under both protocols (0.26/0.08).

Binary Map. Collapsing continuous risk to a single bit (risk = 1, no risk = 0) erases ordinal relationships among threat levels. The resulting rates (ST-P3 Avg. 0.48; UniAD Avg. 0.24) are far worse than RiskOcc (0.26/0.08) and even degrade over the no-map baseline (0.40/0.17), showing that a gradient-free binary signal misleads the planner. Hence risk stratification, rather than mere spatial occupancy, constitutes the core mechanism behind the safety gains.

Gaussian Map. As a control, a random Gaussian map of identical size but carrying no risk semantics yields the worst rates across all metrics (ST-P3 Avg. 0.55; UniAD Avg. 0.33), below the no-map baseline, ruling out gains from the added cross-attention parameters and confirming that RiskOcc’s benefit stems from semantically meaningful risk encoding.

Together, these ablations rule out network capacity, high-dimensional semantics, and binary occupancy as alternative explanations, providing evidence consistent with—though not a direct measurement of—the conditional mutual information $\Delta I > 0$ in Eq. (4).

Planning Performance ofROIDrive

Overall performance. On the nuScenes (Caesar et al. 2020) validation set,ROIDrive attains an average L2 error and collision rate of 0.35 m / 0.08% under the UniAD (Hu et al. 2023) protocol and 0.69 m / 0.33% under the ST-P3 (Hu et al. 2022) protocol, with the lowest collision rate under the UniAD protocol among competing methods.

Table 5:ROIDrive ablation 2 on the RiskOcc-only model (without ego-token initialization or TQC). C1: Bi-Mamba2; C2: SDMap+ Fusion.

Components		UniAD Collision Rate ↓				ST-P3 Collision Rate ↓			
C1	C2	1 s	2 s	3 s	Avg.	1 s	2 s	3 s	Avg.
✓		0.08	0.11	0.30	0.16	0.08	0.18	0.84	0.37
	✓	0.05	0.12	0.22	0.13	0.06	0.27	0.54	0.29
✓	✓	0.08	0.07	0.21	0.12	0.07	0.07	0.64	0.26

Ablation: framework modules. Table 4 incrementally adds the modules. RiskOcc alone (M1) sets a dense risk prior and a safe planning baseline; ego-token initialization (M2) adds the ego motion prior and sharply lowers average L2 (UniAD 0.52 → 0.34, ST-P3 0.92 → 0.67) at marginal collision cost; TQC (M3) then reduces collisions relative to the ego-initialized variant (UniAD 0.15% → 0.08%, ST-P3 0.36% → 0.33%) by down-weighting unreliable queries while keeping L2 nearly unchanged, achieving the intended safety-accuracy balance.

Ablation: occupancy components. Built on the RiskOcc-only model without ego-token initialization or TQC, Table 5 isolates the two components of the branch. The Bi-Mamba2 backbone (C1) provides dynamic avoidance from onboard vision, whereas the SDMap+ prior (C2) fills static blind spots and most reduces the 3 s collision rate; enabling both yields the lowest average collision rate (UniAD 0.12%, ST-P3 0.26%), confirming that dynamic perception and static priors are complementary.

Conclusion

This paper introduced risk-aware occupancy as a new perception representation for E2E autonomous driving. We constructed RiskOcc4D-nuScenes, fully compatible with Occ3D-nuScenes, under a standardized multi-level eval-

uation protocol. We validated our 3D occupancy methods and demonstrated their compatibility and transferability. By fusing the predicted risk-aware occupancy map into GenAD (Zheng et al. 2024) via sliding-window sampling, we yielded collision-rate reductions of 52.9% (UniAD) and 35.0% (ST-P3). We further designed the ROI Drive, an E2E framework coupling an independent risk-aware occupancy branch with temporal query consistency for safety-oriented planning.

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